Solution Requirements

Date	16 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system allows users to securely log in and access the HDI prediction application.	High
2	Authorization levels	Authorized users can enter development indicators and view HDI prediction results.	High
3	External interfaces	The system interacts with the HDI dataset and trained Machine Learning model for prediction.	High

4	Transactions processing	The system validates user inputs, processes the data, and generates HDI predictions.	High
5	Reporting	The system displays the predicted HDI score, development category, and analysis results.	High
6	Business rules, etc.	HDI prediction is based on Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per capita.	High
7	Compliance to laws or regulations	The system ensures secure handling of user data and follows basic data privacy guidelines.	Medium
8	*Other*	The application provides a simple, responsive, and user-friendly interface for easy HDI prediction.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
1	Performance & Speed	The system should process user inputs and generate HDI predictions quickly without noticeable delay.	Response time < 2 seconds

2	Scalability	The system should support an increasing number of users and larger HDI datasets efficiently.	Supports 1000+ users
3	Security & Data Privacy	User input data should be securely handled and protected from unauthorized access.	Secure data handling and access control
4	Reliability & Availability	The system should provide consistent predictions and remain available whenever users access it.	99% system uptime
5	Usability & Accessibility	The application should have a simple, responsive, and easy-to-use interface for all users.	User-friendly interface with easy navigation
6	*Other*	The system should be easy to maintain, update, and enhance with new features in the future.	Easy maintenance and future scalability